

Exercice 1:

Given:

$$B = 3100 \text{ Hz}$$

$$\text{SNR} = 30 \text{ dB} \rightarrow 10^{\frac{30}{10}} = 10^3 = 1000$$

$$C = B \log_2 (1 + \text{SNR})$$

$$= 3100 \log_2 (1 + 1000)$$

$$= 3100 \log_2 (1001)$$

$$= (3100)(3.32) \log_{10} (1001) \quad \leftarrow \log_2 x = \frac{\log_{10} x}{\log_{10} 2}$$

$$= (3100)(3.32)(3)$$

$$= 30880 \text{ bps}$$

$$\approx \boxed{31 \text{ kbps}}$$

↓

This exceeds the max. limit given by

$$C = 2B = 2(3100) = \boxed{6200 \text{ bps}}$$

∴ to achieve 31000 bps, we must use N signal levels

from $C = 2B \log_2 N$

$$\log_2 N = \frac{C}{2B} = \frac{31000}{2(3100)} = 5$$

$$\log_2 N = 5$$

$$N = \text{ans! } \log_2 5$$

$$N = 2^5 = 32$$

$$\boxed{N = 32} \text{ signaling levels}$$

Exercice 2:

Given

$$B = 12.5 \text{ KHz} = 12500 \text{ Hz}$$

$$\text{SNR} = 25 \text{ dB} = 10^{\frac{25}{10}} = 316.2$$

$$a, \quad C = 2B = 2(12500) = \boxed{25000 \text{ bps}}$$

$$\begin{aligned} b, \quad C &= B \log_2 (1 + \text{SNR}) \\ &= (12500) 3.32 \log_{10} (1 + 316.2) \\ &= (12500) (3.32) (2.5) \\ &= \boxed{103805.33 \text{ bps}} \end{aligned}$$

$$c, \quad C = 2B \log_2 N$$

$$\log_2 N = \frac{C}{2B} = \frac{103805.33}{2(12500)}$$

$$= 4.15$$

$$N = 2^{4.15} = 17.75$$

$$\boxed{N = 17} \text{ signaling levels}$$

Exercise 3:

Given

* Number of errors in 1 sec. (n)

$$n = 5$$

* bit transmission rate (b)

$$b = 10 \text{ Mbps} = 10 \times 10^6 \text{ bps}$$

* Bit Error Rate (BER)

$$\text{BER} = \frac{n}{b} = \frac{5}{10 \times 10^6} = \frac{5}{10^7} = 5 \times 10^{-7}$$

$$\boxed{\text{BER} = 5 \times 10^{-7}}$$

Exercise 4:

Given:

Character 1: 1011001

Character 2: 0010100

Odd parity: Character 1 $\rightarrow$ 1011001 $\boxed{1}$
Character 2 $\rightarrow$ 0010100 $\boxed{1}$

Even parity: Character 1 $\rightarrow$ 1011001 $\boxed{0}$
Character 2 $\rightarrow$ 0010100 $\boxed{0}$

Exercise 5:

Given:

data 1101001

generating constant 101.

$$\frac{M(x)}{G(x)} = Q(x) + R(x)$$

XOR		
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

$G(x) = x^2 + 1$ since $101 = x^2 + 0 + x^0$
 $\uparrow$ degree 2 so we append 00 to data.

$M(x) = 110100100$

$\frac{M(x)}{G(x)}$ division is accomplished by successive XOR.
 we ignore the quotient $Q(x)$ and we take the remainder $R(x)$ only.

step 1: 110 100 100

XOR $\begin{array}{r} 110 \\ 101 \\ \hline 011 \end{array} \rightarrow$ this replaces the first 3-bits begin with 1.
 011 100 100

XOR $\begin{array}{r} 111 \\ 101 \\ \hline 010 \end{array} \quad 001000100$

XOR $\begin{array}{r} 100 \\ 101 \\ \hline 001 \end{array} \quad 000010100$

XOR $\begin{array}{r} 101 \\ 101 \\ \hline 000 \end{array} \quad 000000000 \rightarrow$ Remainder since $G(x)$ is degree 2.

$$\text{Therefore CRC} = \text{Data} + R(x) \\ = \boxed{110100100}$$

Exercise 6:

Given:

Data character: 01101010

Hamming Code position: ^{12 11 10 9 8 7 6 5 4 3 2 1} H 0 1 H 1 0 H 1 0 H 1 0

$$m = 8$$

$$n = 4$$

(a)

$$2^n \geq m + n + 1$$

m = number of data bits

n = number of bits in Hamming Code

$$2^4 \geq 8 + 4 + 1$$

$$\boxed{16 \geq 13} \quad \checkmark$$

(b)

Data bits are "1" at positions 2, 5, 8 and 10.

$$2 = 0010, \quad 5 = 0101, \quad 8 = 1000, \quad 10 = 1010$$

$$\text{position code 2} \rightarrow 0010$$

$$\text{position code 5} \rightarrow 0101 \text{ XOR } 0111$$

$$\text{position code 8} \rightarrow 1000 \text{ XOR } 0111$$

$$1111$$

$$\text{position code 10} \rightarrow 1111$$

$$1010$$

$$0101$$



$$\text{Hamming Code bits} = \boxed{0101} \text{ left-to-right.}$$

(c) 12-bits transmitters code.

0 0 1 1 1 0 0 1 0 1 1 0

0 1 0 1 $\rightarrow$ Hamming Code bits.

* Receiver knows the Hamming bits (0 1 0 1)

Assume an error at position 11 (0 received as 1).

* The R_x XOR's the code with bit number of each position in the word containing 1.

$R_x =$ ^{12 11 10 9 8 7 6 5 4 3 2 1}
0 1 1 1 0 0 1 0 1 1 0 $\leftarrow$ bit position.

From Right to Left.

0101	$\leftarrow$ Hamming Code
XOR 0010	$\leftarrow$ position 2.
<u>0111</u>	
XOR 0101	$\leftarrow$ position 5
<u>0010</u>	
1000	$\leftarrow$ position 8
XOR 1010	
<u>1010</u>	$\leftarrow$ position 10
XOR 0000	
<u>1011</u>	$\leftarrow$ position 11.
XOR 1011	

1011 = (11) $\leftarrow$ indicates an error at position 11.